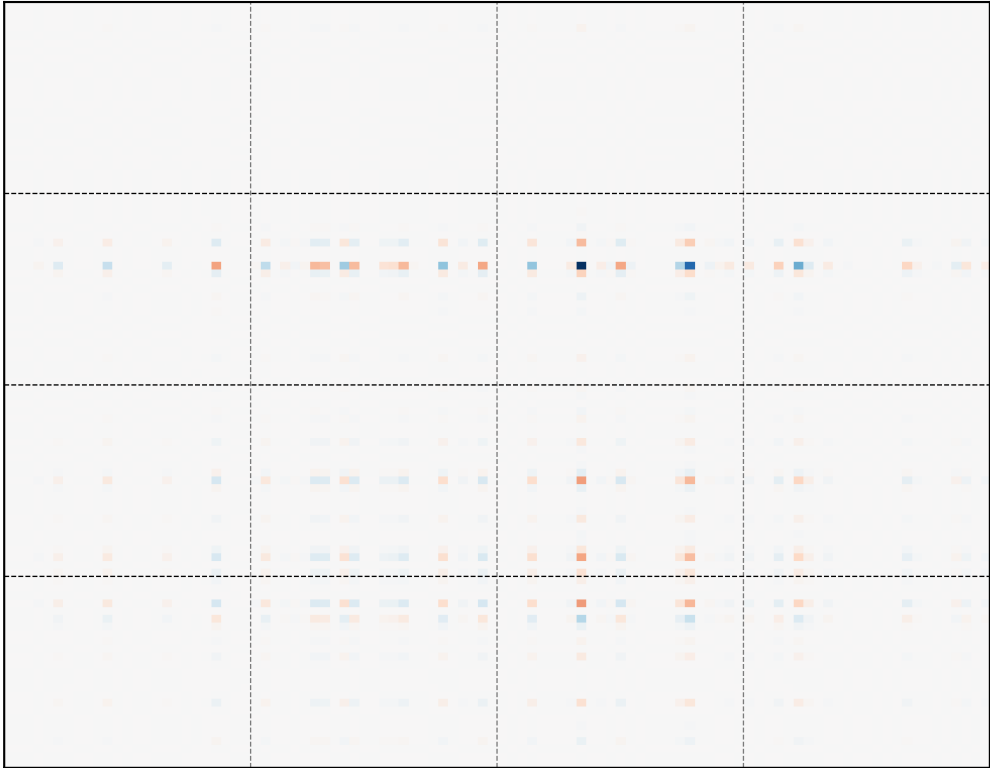
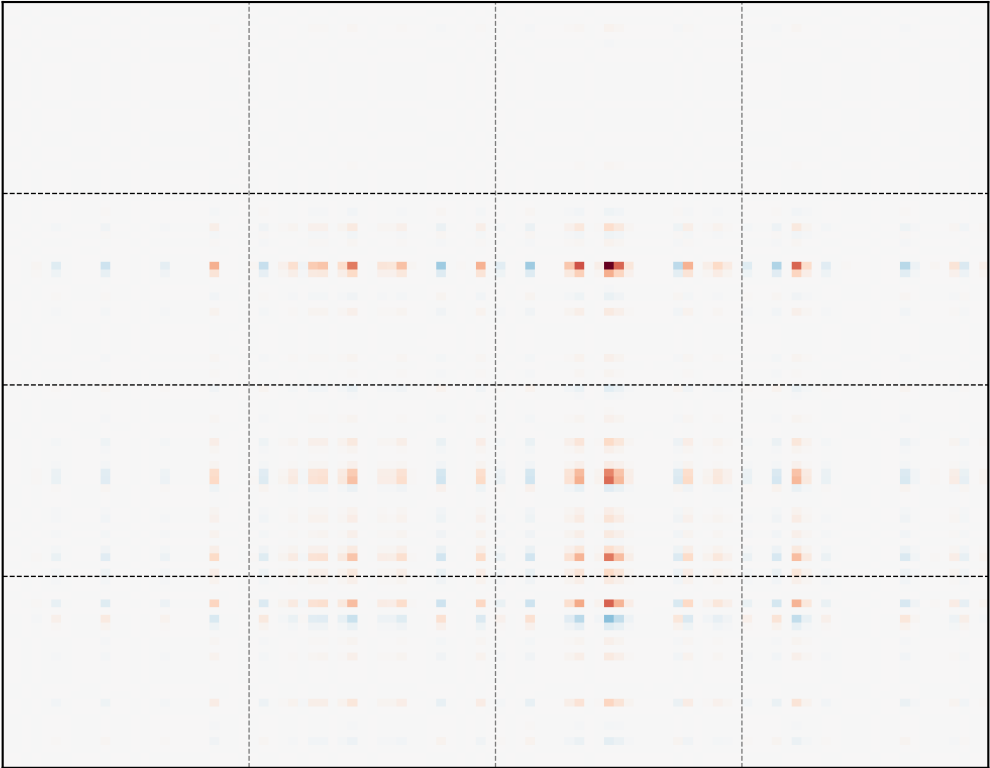


Figure A2: Multi-Scale Decomposition — Biological Module Extraction (BRCA)
Top 6 rank-1 components across 3 scales reveal distinct cancer pathways

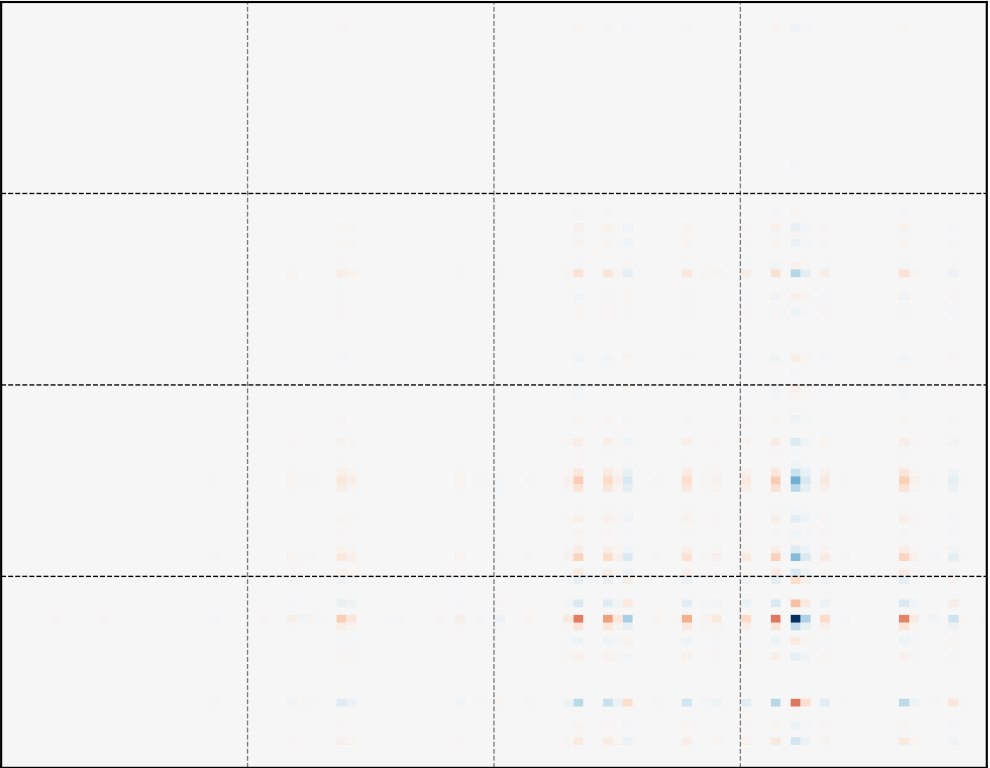
**Estrogen
Pathway
(sigma=3.100)**



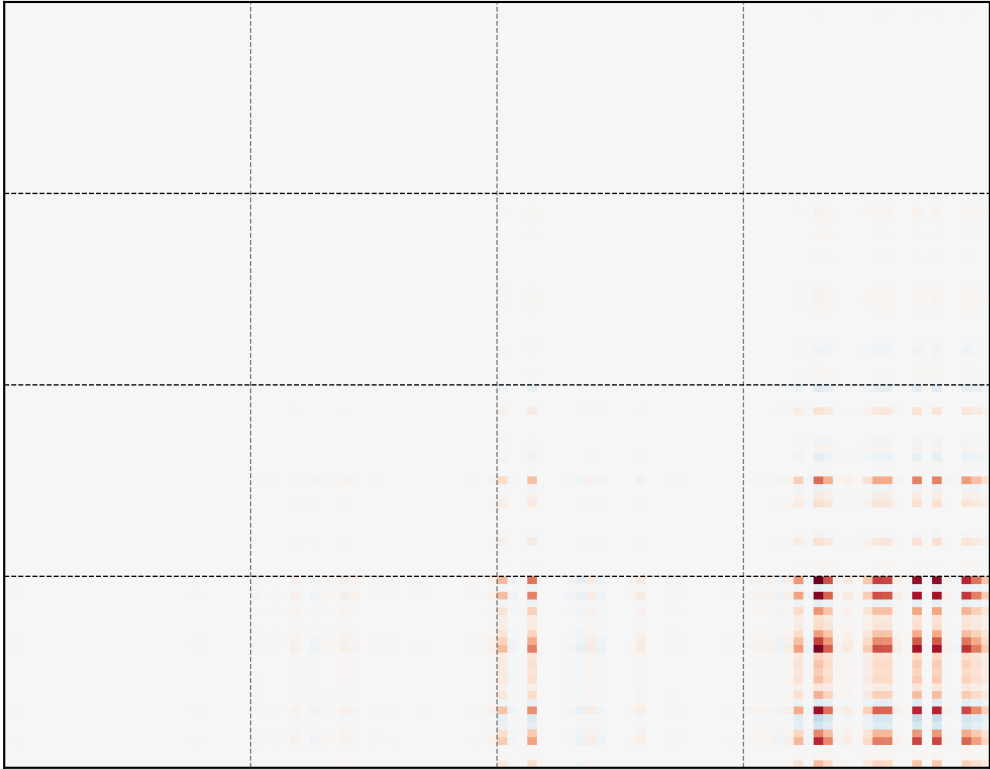
**Cell Cycle
Regulation
(sigma=2.796)**



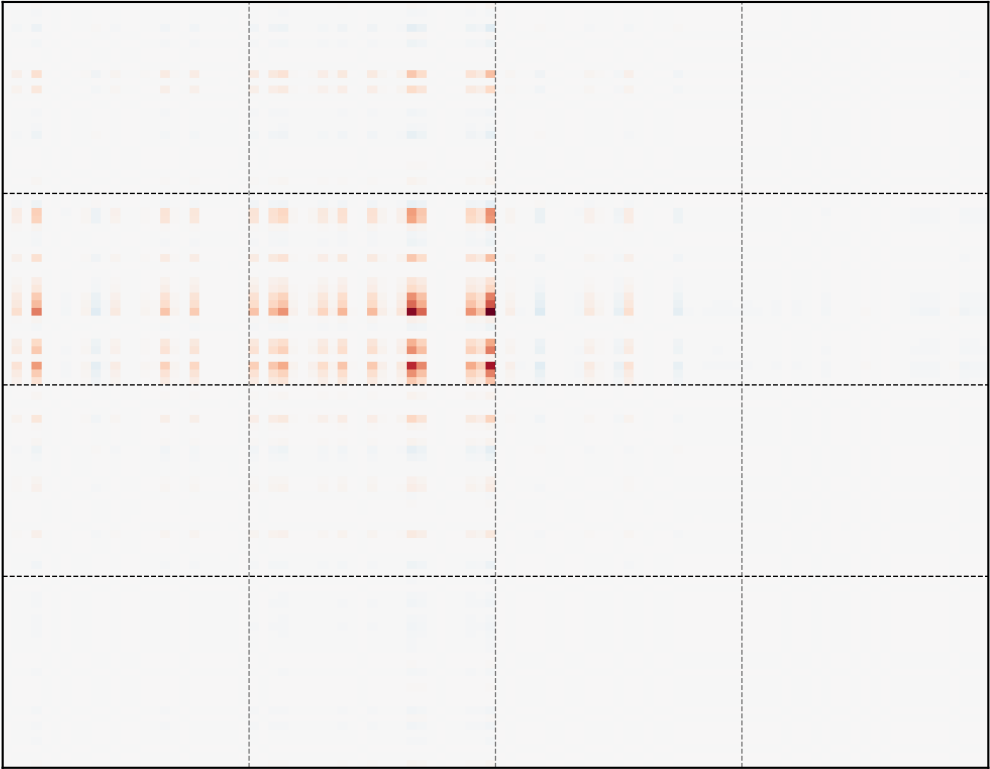
**Immune
Evasion
(sigma=2.539)**



**Metabolic
Reprogramming
(sigma=5.187)**



**Cross-Module
Spillover
(sigma=4.384)**



**Global
Coordination
(sigma=3.762)**

